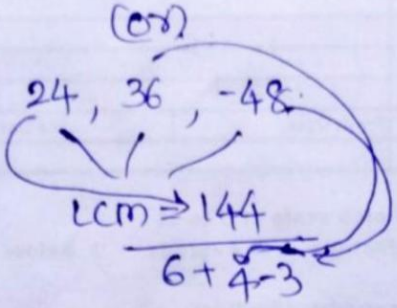
(67)

Total time together = $\frac{1}{\text{Comb eff}} = \frac{144}{7} = 20\frac{4}{7}$ hrs

$\therefore 20\frac{4}{7}$ hr's to fill the tank.

Result Qs 'tve' means filling, -ve means emptying.

(or)



$= \frac{144}{7} = 20\frac{4}{7}$ hrs. ⇒

Questions [Time & Work]

Ex-8:- [B.Q:-1]:- If 9 men working 6 hr's a day can do a work in 88 days. Then 6 men working 8 hr's a day can do in how many days?

Ans:- here work is same in both the cases.

$\therefore M_1 D_1 H_1 = M_2 D_2 H_2$

$9 \times 88 \times 6 = 6 \times D_2 \times 8 \Rightarrow \boxed{D_2 = 99 \text{ days}}$

Ex:-9:- [B.Q:2]:- If 34 men completed 2/5th of work in 8 days working 9 hrs a day. How many more men should be engaged to complete the rest in 6 days working 9 hrs a day?

Ans:- Remaining work = $W_2 = (1 - 2/5) = 3/5$, $W_1 = 2/5$.

$M_1 = 34$, $D_1 = 8$, $H_1 = 9$, $D_2 = 6$, $H_2 = 9$, $M_2 = (34 + \text{extra men}(x)) = 34 + x$

Assume extra men = x

$\frac{W_1}{M_1 D_1 H_1} = \frac{W_2}{M_2 D_2 H_2}$

$\Rightarrow \frac{2}{34 \times 8 \times 9} = \frac{3}{(34+x) \times 8 \times 9} \Rightarrow 34 + x = 34 \times 2 = 68$

$\Rightarrow \boxed{x = 34}$

Ex:-10:- [B.Q:-3]:- If 5W (or) 8G can do a work in 84 days. In how many days can 10 women & 5 girls can do the same?

Ans:- $5W = 8G = \frac{1}{84} \Rightarrow 1W = \frac{1}{84\times5}$, $1G = \frac{1}{84\times8}$

$\Rightarrow 10W \& 5G \Rightarrow 10W + 5G = 2(5W) + 5G$

$= 2\left(\frac{1}{84}\right) + 5\left(\frac{1}{8\times84}\right) = \frac{8\times2+5}{8\times84} = \frac{21}{8\times84_4}$

$\Rightarrow \frac{1}{32}$ = Together efficiency

days = 1/Tog eff = 32 days.

Ex:-11:- [B.Q:-5]:- $A = \frac{1}{18}$, B can do the same in half time of A. Then working together, what part of the work they can finish in a day?

A can do in 18 days $\Rightarrow A = \frac{1}{18}$.

'B' can take half the time of A means $\frac{18}{2} = 9$ days.

$\Rightarrow B = \frac{1}{9}$

together 1 day work = Combined eff $= \frac{1}{18} + \frac{1}{9} = \frac{3}{18} = \frac{1}{6}$

$\therefore \frac{1}{6}$th of work they can in one day.

Ex:-12:- [B.Q:-6]:- A is twice eff/as good as B. & together they finish in 18 days. Then in how many 'A' alone can finish the same?

Ans:- $A = 2B$ & B in 18 days. & 'A' alone in = ?

↓ More eff person.

$\therefore$ T.N.D together = $\frac{\text{No. of times} \times \text{More eff person days}}{1 + \text{no. of times}}$

$18 = \frac{2 \times \text{More eff person days}}{1+2} = \frac{18\times3}{2}$

$\Rightarrow \frac{18\times3}{2}$ = More eff person days = 27 days.

Ex:13:- [B.Q:-8]:- A can do a certain work in 7 days of 9 hrs. each & B alone can do it in 6 days of 7 hrs each. How long will they take to do it working together 8(2/5) hrs a day?

Ans:- A can take (7 days x 9 hrs a day) 63 hrs to complete

B " " (6 days x 7 hrs a day) 42 hrs to complete

No. of days by 'A' if $8\frac{2}{5}$ hrs a day working $= \frac{63}{8\frac{2}{5}}$

$= \frac{63}{42/5} = 63 \times \frac{5}{42} = \frac{45}{6}$ days.

No. of days by 'B' if $8\frac{2}{5}$ hrs a day working $= \frac{42}{8\frac{2}{5}}$

$= \frac{42}{42/5} = 42 \times \frac{5}{42} = 5$ days.

Together $= \frac{xy}{x+y} = \frac{\frac{45}{6} \times 5}{\frac{45}{6}+5} = \frac{45 \times 5/6}{45+30/6} = \frac{225}{75} =$ **3 days**

(Or)

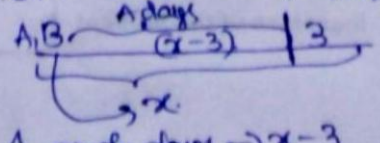
A $\longrightarrow$ 63 hrs, B $\longrightarrow$ 42 hrs.

together hrs $= \frac{63 \times 42}{63+42} = \frac{63 \times 42}{105}$ hrs

No. of days when $8\frac{2}{5}$ hrs per day working $= \frac{\frac{63 \times 42}{105}}{8\frac{2}{5}}$

$= \frac{\frac{63 \times 42}{105}}{42/5} = \frac{63 \times 42}{105} \times \frac{5}{42} = 3$ days

Ex:14:- [B.Q:-9]:- $A = \frac{1}{14}$, $B = \frac{1}{21}$, A & B started the work but 3 days before completion A left. Total no. of work done days?



A no. of days $\rightarrow x-3$

B " " " $\rightarrow x$

$mA + nB = 1$

$(x-3)\frac{1}{14} + x \cdot \frac{1}{21} = 1 \Rightarrow \frac{3(x-3)+2x}{42} = 1$

$\Rightarrow 3x - 9 + 2x = 42 \Rightarrow 5x = 51$

$\Rightarrow \boxed{x = \frac{51}{5} = 10\frac{1}{5} \text{ days}}$